

AI Textbook Q&A System

A RAG-Based Educational Question Answering System with
Deep Source Tracing

CST8507: Natural Language Processing

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Introduction

Problem:

Students studying AI/ML face challenges navigating multiple textbooks to find relevant information.

Solution:

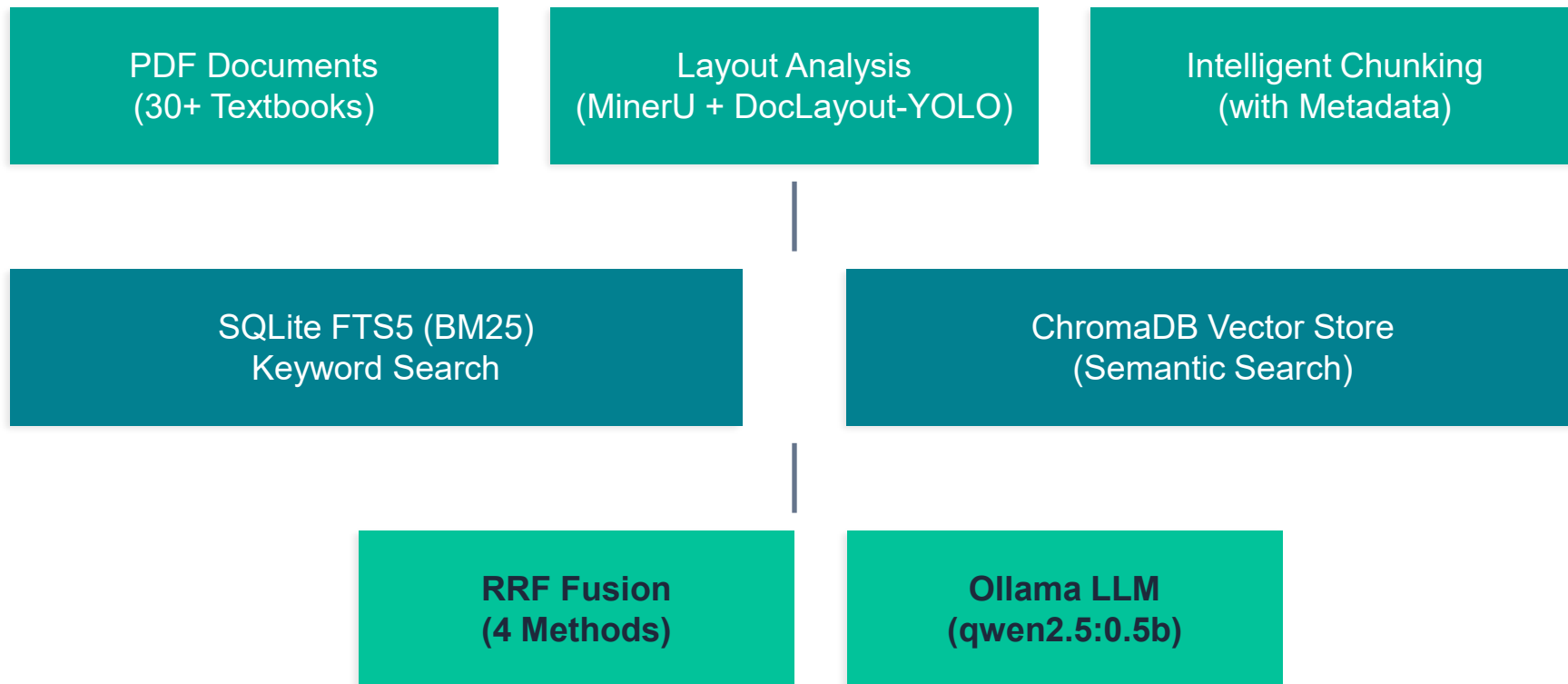
A RAG-based Q&A system that answers educational questions using 30+ canonical AI/ML textbooks.

Key Innovation:

Deep Source Tracing — not just citing books, but pinpointing exact pages with highlighted regions.

Research Question: How can we build a transparent, traceable educational Q&A system for AI/ML learning?

System Architecture



Hybrid Retrieval Architecture

1. SQLite FTS5

BM25 keyword search for fast exact matching

2. ChromaDB Vector

Semantic search with sentence-transformers embeddings

3. PageIndex Tree

Hierarchical TOC navigation using LLM reasoning

4. Metadata Filter

Structured filtering by book, chapter, page

Reciprocal Rank Fusion (RRF) combines all

Dataset

Domain	Books	Examples
Machine Learning	7	ISLR, ESL, PRML, Deep Learning
Mathematics	5	MML, Convex Optimization
NLP	3	SLP3, Intro to IR
Reinforcement Learning	1	RL: An Introduction
Computer Vision	1	Computer Vision (Szeliski)
Python	3	Fluent Python, Python Cookbook

30+

Textbooks

~500MB

PDF Documents

Open Access

Educational Use

Part 1: Web Interface

Streamlit-based UI with Deep Source Tracing

Question Input

Natural language text box for user queries

System Answer

AI-generated response based on retrieved documents

Source Documents

Clickable references showing exact book, page, and region

PDF Viewer

Original PDF page with highlighted source region

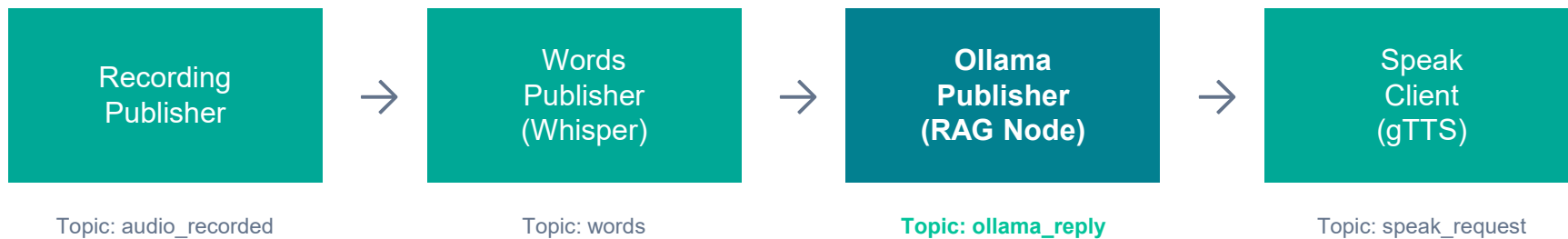
Local URL

<http://localhost:5173>

Screenshots of the web interface are
available in the project folder

Part 2: ROS2 Integration

Vocal Interaction Pipeline



ROS2 Terminal Screenshots (screenshot2.png & screenshot3.png)

Evaluation

20

Test Questions

>80%

Target Accuracy

3

Retrieved Chunks

Scoring Criteria

Score	Meaning
1.0	Correct - Answer fully addresses the question
0.5	Partially Correct - Answer partially addresses the question
0.0	Incorrect - Answer does not address the question

Results

Key Achievements

- Successfully implemented hybrid RAG architecture with 4 retrieval methods
- Deep source tracing with clickable PDF references and highlighted regions
- Layout-aware document processing preserving tables, formulas, and figures
- Full ROS2 integration with vocal interaction pipeline
- Low memory footprint (<1.5GB) suitable for robot deployment

Tech Stack: Python | Ollama | ChromaDB | SQLite | Streamlit | ROS2 | MinerU | Whisper | gTTS

Challenges & Solutions

Challenge 1

Complex PDF layouts with tables and formulas

Solution:

Used MinerU with DocLayout-YOLO for layout-aware parsing

Challenge 2

Balancing retrieval accuracy with speed

Solution:

Implemented RRF fusion combining keyword and semantic search

Challenge 3

Memory constraints for robot deployment

Solution:

Selected qwen2.5:0.5b model (~0.4GB footprint)

Future Work

- Multi-language support
- Real-time collaborative Q&A
- Mobile app integration
- Expanded knowledge domains

Conclusion

We built a RAG-based educational Q&A system that:

- Answers AI/ML questions using 30+ canonical textbooks
- Provides deep source tracing to exact PDF pages
- Integrates seamlessly with ROS2 for robot deployment
- Runs locally with low memory footprint

Thank You!